Electric history 1a

Read the information below and then answer the following questions about text and language.

(1) It was a long time ago that Thales of Miletus (Greece, 625–550 BC) first experimented with amber. Amber is fossilised tree sap, usually occurring in small lumps, which are translucent and golden. (Still today amber is made into jewellery.) The Greek word for amber is *'elektron'*. In 530 BC Thales noticed that little pieces of sawdust were attracted and then stuck to a piece of amber that had been rubbed with a cloth. On the other side of the globe Chinese investigators (300 BC) were experimenting with a material called magnetite. Curved magnetite spoons, which were able to spin on a bronze platform, always pointed in the same direction. Eventually this became the magnetic compass in Europe. Although the attraction of amber and magnetite seemed to be similar, it was not clear what attracted the magnetite, or why rubbed amber attracted the sawdust. Finally, in 1750, the Italian scientist Girolama Cardano (1501–1576) performed experiments which showed that amber and magnetite were in fact quite different: Cardano was distinguishing between electrostatic and magnetic attraction.

(2) In 1729 the English scientist Stephen Gray (1666–1736) (discover / discovers / discovered) that electricity (must move / could move / has moved) from one place to another. He (try / tries / tried) to send electricity from a charged object through all kinds of materials and (find / finds / found) that some materials allowed electricity to (flowed / flow / flew) and others (do / does / did) not. Gray (call / called / calls) them conductors and insulators. Gray also (state / states / stated) that electricity (is / was / were) a fluid that covered the outer surface of an insulator. In 1733 Charles-Francois de Cisternay Dufay (France, 1698–1739) (polish / polished / polishing) glass spheres and resin spheres with a cloth. He learns / learnt / learned that charged resin spheres (repealed / repells / repelled) each other, as (do / did / done) charged glass spheres, but charged resin spheres (attract / attracted / attracts) charged glass spheres. Dufay (conclude / concluded / has concluded) that there (are / was / were) two different fluids of electricity. After (conducting / having conducted / has conducted) many electrical experiments in 1740, Benjamin Franklin (Philadelphia, 1706–1790) (believe / believed / believing) that there (is / are / was) only one kind of fluid. He (defined / define / defines) too much electric fluid as positive electricity and too little electric fluid as negative electricity. He also (stated / did state / states) that the electric fluid (flown / flowing / flowed) from positive to negative. However, today we believe that too much electricity is an excess of negatively charged electrons and that an electric current moves electrons from negative to positive. As a result, there is some confusion about the direction of electricity today.

(3) In 1827, German scientist Georg Ohm (1789–1854) was intrigued by the results when he measured electricity. He found that electric pressure measured in volts, electric current measured in amperes and electric resistance measured in ohms always behaved in a certain pattern. This relationship is now known as 'Ohm's Law'.

Looking at the information

1 Write a brief (one sentence) summary of each of the three paragraphs above.

2 Complete the timeline below by indicating the dates and names of the scientific contributions mentioned in the text.



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Electric history

1b

3 Summarise the contribution of each scientist in a table. Include some detail about the scientists.

Name	Date	Country	Contribution

4 Identify the scientist who lived the longest and the scientist who had the shortest life. State their ages.

Looking at the language

5 Underline the verb groups in paragraph 1. Three examples have been done for you already.

6 Select the correct verb form from each selection in paragraph 2 and underline it. One example has been done for you. Then rewrite the passage using only the correct verbs.

7 There are four main types of verbs:

- **action verbs** describe events;
- **linking verbs** join characteristics to things;
- **saying verbs** describe how something is said;
- **thinking verbs** show feelings and thoughts.

The text above is an historical recount, which usually has all four types of verbs. Complete the table below by identifying three examples for each verb type. One example is already given.

Action verb	Linking verb	Saying verb	Thinking verb
build	is made up of	shout	feel

8 In an *active voice*, the person performig the action is placed at the front of the sentence. In *passive voice* that person is placed at the back of the sentence, or left out altogether.

Examples: The scientist measured the electric current. [ACTIVE VOICE]
The electric current was measured (by the scientist). [PASSIVE VOICE]

Rewrite the first and last sentences of paragraph 3 of the text, changing the voice.

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